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The Sixth–Finger: a Modular Extra-Finger to Enhance Human Hand Capabilities

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***Abstract*— Robotic prosthesis are usually intended as artificial device extensions replacing a missing part of a human body. A new approach regarding robotic limbs is presented here. A modular robot is used not only for replacing a missing part of the body but also as an extra–limb in order to enhance manipulation dexterity and enlarge the workspace of human beings. In this work, the model and control of an additional finger, the Sixth–Finger, is presented as a case study of this type of robotic limbs. The robotic finger has been placed on the wrist opposite to the hand palm. This solution allows to enlarge the hand workspace, increasing the grasp capability of the user. An object-based mapping algorithm is proposed to control the robotic extra-finger by interpreting the whole hand motion in grasping action. A four DoFs modular prototype is presented along with numerical simulations and real experiments. The proposed Sixth–Finger can lead to a wide range of applications in the direction of augmenting human capabilities through wearable robotics.**

I. Introduction

Wearable robotics is a promising way of integration between humans and robots. Typically wearable robot are intended to be a way to substitute missing parts of the human body, by means of prosthesis [1], [2], or to enhance human body force and precision capabilities, by means, for example, of exoskeletons [3], [4].

Recently, robots integrated with humans have been actively developed as wearable robots by taking advantage of the progress in miniaturization and efficiency of the technological components. The purpose of these new devices is not to enhance the lift strength far above human capability by wearing a bulky robot, but to support human capability within its range by wearing lightweight and compact robots [5]. As example, in [6] the authors presented the Supernumerary Robotic Limbs (SRL), a wearable robot designed to assist human with additional arms attached to the wearer's body. Such devices, further investigated in [7] and [8] can closely cooperate with the human, for instance holding an object, lifting a weight, positioning a work piece, etc..

It is very interesting to observe how these wearable robots are potentially able to enhance the strength and the precision of the human users, and also to enrich the skills and tasks that can be performed. In such work the authors furthermore observe that the wearer and the extra–arms work very closely that he/she could potentially perceive them to be his/her own.

In this paper we propose to further exploit this approach. While the works previously described are relative to devices that can be applied as extra–arms or extra–limbs, we focus on a smaller but paramount part of our body: the hand. The goal is then to integrate the human hand with an additional robotic finger: the Sixth–Finger (Fig. 1).

Fig. 1. The first prototype of the Sixth–Finger.

The challenge is that of designing a new finger which plays the role of an additional thumb [9]. The thumb is the finger which lets us manipulate objects in a complex way [10]. The main function of the human thumb is the pincer action that allows to grasp objects, to hold them firmly, to perform complex manipulation (e.g. twisting, rubbing). If we had a second thumb, probably ring and pinkie fingers would be dexterous as the index and middle fingers are. Trained with the second thumb, less-developed fingers will strengthen and gain dexterity. This could give humans the possibility to manipulate objects in a more efficient way, enhancing our hand grasping dexterity/ability. As an extension of the human hand, the second thumb could make certain tasks much easier, and extend our hand's workspace.

The aim of the prototype described in this work is to make human hand more symmetric, so that when the human fingers close (when phalanxes flex), the extra–finger reflects exactly these motions. This extension is going to increase the human hand workspace and its grasping abilities. The difficulties in achieving such improvements lie both in the design and the control of the Sixth–Finger device. For the design aspect, the extra–thumb has to be sufficiently small and light in order to be easily worn and carried by the user. We developed a modular finger that can be worn on the wrist with the help of a rubber band. The structure of the modules is obtained by using rapid prototyping techniques, while the active degrees of freedom (DoF) are realized with servomotors. Concerning the control, we decided to capture the motion of the human hand through a dataglove and map such motion on the extra–finger. This solution differs from classical techniques for exoskeleton controllers which are based on the reading of some bio-signal like EMG [11]. Our approach goes in the

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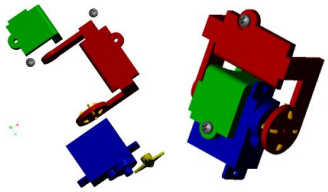

Fig. 2. The exploded view of a module. The blue part is the servo motor, while the green and red parts form the module structure.

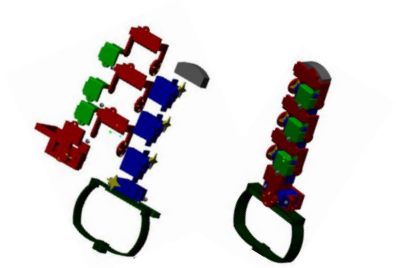

Fig. 3. The Sixth-Finger exploded and normal view. Three single–DoF modules are connected through a fourth servomotor to a wrist rubber band. The fourth servomotor accounts for abduction/adduction motion while the three modules accounts for flexion/extension motion.

direction of smart devices able to interpret human action and move accordingly. In this paper we present an object-based mapping algorithm, to map hand motion onto the Sixth–Finger. The algorithm extends the method proposed in [12] to the case of a human hand augmented with a robotic extra–finger. The paper is organized as it follows. In Section II the modular design of the Sixth–Finger is detailed. Section III deals with the mapping algorithm used to control the robotic device, while in Section IV some numerical simulations proving the applicability of the mapping are reported. Finally in Section V experiments with the first prototype of the extra–finger are presented and in Section VI conclusion and future work are outlined.

II. Modular Design of the Sixth Finger

The Sixth-Finger has been designed and developed considering a modular structure. The modular approach allows the usage of only the necessary number of degrees of freedom to accomplish a desired grasp [13]. Great advantages are obtained in versatility since the extra-finger can be disassembled and reassembled to form new morphologies suitable for new tasks. Modularity also offers robustness considering that robot parts are interchangeable [14]. Each module is composed of a servo motor and two plastic connections which define the module structure. As a result, the module has one DoF and the following dimensions: $42 \times 33 \times 16$mm. In Fig. 2 the exploded and normal view of the module are presented. The modules can be easily connected through screws to obtain more complex kinematic chains like the modular hand described in [15]. The Sixth-Finger is a four DoFs modular structure. Three DoFs are obtained considering three modules in a pitch–pitch connection, which is used to replicate the flexion/extension motion of the fingers. The modules are connected through a fourth servo motor to a rubber band that allows to wear the device on the wrist. The fourth servo motor reproduces the abduction/adduction capability of the finger. The complete structure of the Sixth-Finger is reported in Fig. 3.

Table I shows the Denavit Hartenberg parameters of the modular device. All the motors are PWM controlled through an Arduino Uno board [16]. This solution guarantees flexibility in the control strategies and allows to interface the extra–finger with other device for hand tracking as detailed in Section V.

TABLE I

Denavit–Hartenberg parameters for the Sixth–Finger. Parameter a is the same for all the phalanges due to the modular structure of the finger, and its value is $a = 42$ mm.

Joint	id.	a_j	α_j	d_j	θ_j
MCP	1	0	$-\pi/2$	0	θ_1
MCP	2	a	0	0	θ_2
PIP	3	a	0	0	θ_3
DIP	4	a	0	0	θ_4

III. Modeling and Controlling the Extra–Finger

In this section we describe the general procedure proposed to control extra–finger joints. We will refer to an arbitrary number of extra–fingers to underline the generality of the approach. Let us define a reference frame Σ_0 on the hand. Its origin, O is in the wrist center of rotation, the z axis is perpendicular to hand palm plane, the x axis is the intersection between the sagittal and the transverse hand plane, pointing towards the little finger, the y axis is consequently defined [17].

Let us indicate with $p_i^h \in \Re^3$, $i = 1, ..., n_h$ the coordinates of reference points on the reference human hand, expressed w.r.t. Σ_0. Although the mapping proposed in [12] is general and does not require any constraints in the choice of the reference points, in this paper we choose the five fingertips of the human hand, therefore $n_h = 5$. We assume that the position of such reference points can be measured or evaluated, for instance by means of an instrumented glove. Let us define as augmented hand the system composed of the hand with its five fingers and the robotic extra–finger devices. On the artificial fingers some reference points are chosen, more specifically, one reference point is chosen at the tip of each artificial finger. The augmented hand is then characterized by a set of reference points that is the union of the set of reference points on the human fingers and the set of points on the artificial ones. The reference points for the augmented hand are then $p_i^a \in \Re^3$, $i = 1, \cdots, n_a$, with $n_a > n_h$. The first n_h reference points are the same of the human hand, i.e. $p_i^h = p_i^a \ \forall i \leq n_h$ while the remaining $n_a - n_h$ points are those relative to the artificial fingers. Let us indicate with $\mathcal{C}$ the minimum bounding sphere defined by the reference points on the augmented hand, i.e. the sphere with the minimum radius containing all the n_a reference

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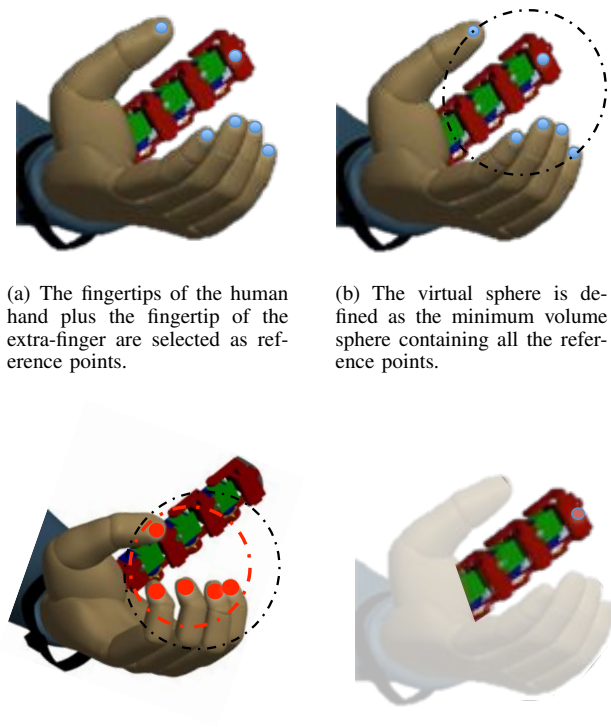

(a) The fingertips of the human hand plus the fingertip of the extra-finger are selected as reference points.

(b) The virtual sphere is defined as the minimum volume sphere containing all the reference points.

(c) The motion of the human hand displaces the reference points and thus deforms and moves the virtual sphere.

(d) The mapping computes the new reference position of the extra-finger.

Fig. 4. The mapping algorithm.

points. Let o^h indicate its center and let r^h be its radius. Let us define a reference frame Σ_1 on the virtual sphere, whose origin is in the sphere center and whose axes are, in the reference starting position, parallel to Σ_0 axes.

When the human hand fingers are moved, the reference point coordinates p_i^h change, let us indicate with $\Delta p_i^h \in \Re^3$, $i = 1, \cdots, n_h$ a vector containing such coordinate variations. Such displacement produces a transformation in the virtual sphere, which we have approximated as the combination of a rigid body motion and an isotropic deformation. The rigid body motion can be furthermore represented as the combination of a translation $\Delta o^h \in \Re^3$ and a rotation $\Delta\Phi \in \Re^3$. The rotation term $\Delta\Phi \in \Re^3$ is defined as

$$\Delta\Phi = [\Delta\phi,\ \Delta\theta,\ \Delta\psi]^{\mathrm{T}},$$

where $\Delta\psi$, $\Delta\theta$ and $\Delta\psi$ represent the rotations angles w.r.t. to x, y and z axes. The non rigid isotropic deformation can be described by the parameter $\Delta s \in \Re$ defined as

$$\Delta s = \frac{\Delta r}{r}.$$

With these definitions, and assuming that the applied displacements are sufficiently small to consider a linear kinematic relationship, we can express the displacement of each reference point on the human hand as

$$\Delta p_i^h = \Delta o^h + \Delta\Phi \times \left(p_i^h - o\right) + \Delta s \left(p_i^h - o\right). \quad (1)$$

It is worth to observe that we approximate the transformation that the hand applies to the virtual sphere as a combination between a rigid body motion, described by the first two terms of the right hand side in eq. (1), and an isotropic transformation (compression or expansion) described by the third term. The third term also account for the sphere radius variation. In this paper we do not consider other types of transformations for the sake of simplicity, however the method can be integrated to include a non–isotropic transformation, as described in [15], and also shear deformations [18]. Eq. (1) can be applied to all the reference points p_i^h, leading to the following linear system

$$\Delta p^h = A\Delta\xi, \quad (2)$$

where $\Delta p^h = \left[\Delta p_1^{h\mathrm{T}}, \cdots, \Delta p_{n_h}^{h\mathrm{T}}\right]^{\mathrm{T}} \in \Re^{3n_h}$ is a vector collecting all the reference point displacements, $\Delta\xi = \left[\Delta o^{\mathrm{T}},\ \Delta\Phi^{\mathrm{T}},\ \Delta s\right]^{\mathrm{T}}$ is a 7×1 vector containing the unknown parameters and $A \in \Re^{3n_h \times 7}$ is the linear system matrix, defined as

$$A = \begin{bmatrix} A_1 \\ \cdots \\ A_{n_h} \end{bmatrix},$$

in which each sub-matrix $A_i \in \Re^{3\times 7}$ is defined as [1] $A_i = \begin{bmatrix} I & -s(p_i^h - o) & (p_i^h - o) \end{bmatrix}$, where $I \in \Re^{3\times 3}$ is the identity matrix. The linear system in eq. (2) can be solved to find

$$\Delta\xi = A^+\Delta p^h + N_A\psi, \quad (3)$$

where A^+ denotes a generic pseudo–inverse of A matrix, while $N_A \in \Re^{7\times h}$ represents a basis of A matrix nullspace, whose dimension is $h \geq 0$, and ψ is an arbitrary h–dimensional vector parametrizing the homogeneous solution of the system. When $h > 0$, the vector ψ can be defined to optimize a cost function that can be selected on the basis of the task. Once the sphere transformation parameters have been evaluated, such transformation can be projected on the reference points on the robotic fingers, i.e, $\forall p_i^a, n_h < i \leq n_a$

$$\Delta p_i^a = \Delta o^h + \Delta\Phi \times (p_i^a - o) + \Delta r\,(p_i^a - o). \quad (4)$$

Let $\Delta p^a = \begin{bmatrix} \Delta p_{n_h+1}^{a\mathrm{T}}, & \cdots, \Delta p_{n_a}^{a\mathrm{T}} \end{bmatrix}^{\mathrm{T}}$ be a $3(n_a - n_h)$–dimensional vector containing the reference point displacements on the artificial fingers. The joints of the artificial fingers are controlled to obtain on the fingertips the displacement evaluated in eq. (4). Let n_q indicate the number of joints of the artificial fingers and let $J_a \in \Re^{3(n_a-n_h)\times n_q}$ indicate their Jacobian matrix [19], we can evaluate finger joint displacements as

$$\Delta q_a = J_a^+\Delta p_i^a + N_J\chi, \quad (5)$$

where J_a^+ denotes a generic pseudoinverse of artificial fingers' Jacobian matrix, $N_J \in \Re^{n_q\times n_\chi}$ is a matrix whose

[1] For any three–dimensional vector $v = [v_1,\ v_2,\ v_3]^{\mathrm{T}}$, we indicate with $s(v)$ the skew matrix associated with vector v, i.e.

$$s(v) = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}.$$

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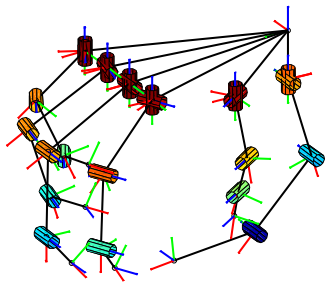

Fig. 5. Scheme of the augmented hand kinematics: links (black lines), reference frames (red, green and blue lines), and joints (cylinders).

columns form a basis of J_a, n_χ represents the dimension of artificial finger redundancy space, $n_\chi \geq 0$, and finally $\chi \in \Re^{n_\chi}$ is a n_χ–dimensional parameter parametrizing the homogeneous part of the solution, which represents the redundant motions of the artificial fingers. Note that while all the reference points of the augmented hand contribute to the definition of the virtual sphere $\mathcal{C}$, only those of the human hand contribute to the computation of the sphere motion and deformation. Moreover, the mapping procedure computes only the new commands for the robotic fingers since the human hand is free to move. The mapping procedure is schematically represented in Fig. 4. In the following section we will show how the procedure works with some numerical experiments.

IV. Numerical Simulations

The performance of the proposed control algorithm have been evaluated through a series of numerical simulations using *SynGrasp* [20]. SynGrasp is a Matlab Toolbox that aims at simulating and analyzing hand grasping. The toolbox includes functions to define the hand kinematic structure and the contact points with a grasped object. New functions have been added to describe hand model where also extra–fingers are considered.

Concerning the human hand, in this work we consider the 20 DoFs hand kinematic model detailed in [21], [22]. The index, middle, ring and pinkie fingers are represented with the same kinematic structure, in terms of joints number and arrangement. The proximal and distal interphalangeal articulations (PIP, DIP) are represented as a one DoF revolute joint, while the metacarpo–phalangeal (MCP) articulations, connecting fingers to the palm, are represented as a two DoFs joint, consisting of two revolute joints with orthogonal axis. In the thumb, the trapezio–metacarpal articulation (TMC) is generally with a 2 DoFs joint with orthogonal axes, while the metacarpo–phalangeal (MCP) and the interphalangeal articulations (IP) are represented as a single DoF revolute joint. The hand dimension, and in particular the length of the bones, is specific to each individual and varies with the age. However, in the literature, there are works on human hand anatomy showing that typically the ratios between the length of each finger can be considered constant [23]. The 20 DoFs structure of the human hand has been augmented with the Sixth–Finger, whose kinematics has been described in Sec. II. The resulting augmented hand structure has then 24 DoFs and is schematically represented in Fig. 5. To control the robotic extra–finger, the mapping algorithm previously illustrated has been implemented.

In the first set of simulations, free hand motions were analyzed. We moved the human hand fingers according to the postural synergies evaluated in [24]. In that work the authors demonstrated that notwithstanding the kinematic complexity of the human hand musculoskeletal structure, a few variables, named postural synergies, are able to describe hand posture during grasps of everyday objects. The role of postural synergies in grasping and manipulation properties was exploited in [21], [22]. The choice of postural synergies is then motivated by the capability of the first two synergies to describe a great amount of hand motions. However, the proposed method is general and can be applied to any kind of hand movements. In the presented simulations, hand joints were activated considering

$$\Delta q^h = S\Delta z,$$

in which $\Delta q^h \in \Re^{n_q}$ represents the displacement of the human hand joints, $S \in \Re^{n_q \times n_z}$ is the synergy matrix, and $\Delta z \in \Re^{n_z}$ represents the synergy activation vector. In the simulations we assumed to activate only the first two synergies, so $n_z = 2$. Functions for the evaluation of direct kinematics available in SynGrasp were adopted to evaluate the corresponding displacements of the fingertips, Δp^h. The procedure detailed in Sec. III was then applied to evaluate the rotations to be applied to the Sixth–Finger, Δq^a. The results obtained are reported in Fig. 6, that shows how the augmented hand moves when the human hand model joints are activated following the first and the second synergies, i.e. choosing $\Delta z = [\Delta z_1,\ 0]^{\mathrm{T}}$ and $\Delta z = [0,\ \Delta z_2]^{\mathrm{T}}$. The first row shows the activation of the first synergy, corresponding qualitatively to the opening–closure movements of the fingers. The second row is relative to the second synergy, that moves the fingertips towards the hand palm. In both cases, as it can be seen, the Sixth–Finger follows the motion of the human fingers.

The second set of numerical simulations was devoted to evaluate the role of the Sixth–Finger in grasping tasks. We considered the grasping of simple shape objects, namely spheres and cubes, with different sizes, varying from 40 mm to 140 mm the radius/edge. For each grasp, we analyzed the values of one of the grasp quality indexes, in this case the *grasp isotropy index* (GII) discussed in [25], defined as the ratio between the minimum $\sigma_{min,G}$ and maximum $\sigma_{max,G}$ singular value of the grasp matrix G,

$$GII = \frac{\sigma_{min,G}}{\sigma_{max,G}}.$$

The index GII measures the contribution of the contact forces to the total wrench exerted on the object, and varies from 0 to 1. Its optimal value, i.e. $i = 1$, corresponds to an isotropic grasp, in which the magnitudes of the internal

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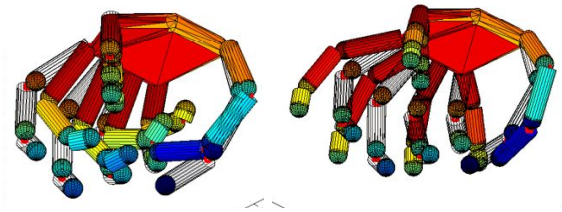
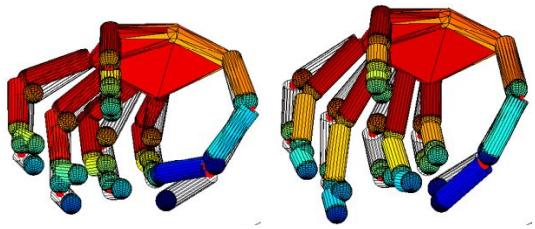
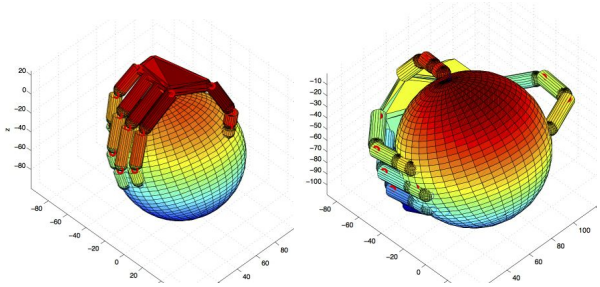


a) First synergy: $\Delta z = [\pm\Delta z_1,\ 0]^{\mathrm{T}}$.

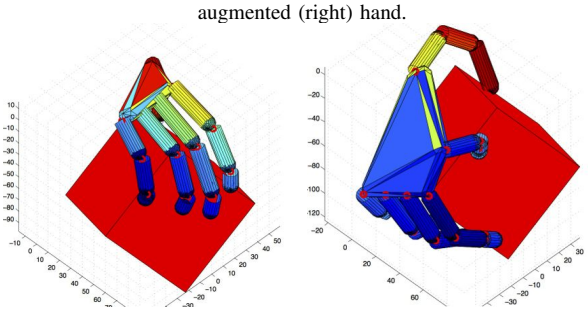


b) Second synergy: $\Delta z = [0,\ \pm\Delta z_2]^{\mathrm{T}}$.

Fig. 6. Activation of the first two human hand synergies. The transparent hand represents the initial position of the augmented hand while the solid the final position. The Sixth Finger is represented in blue. Note that in both the examples the extra–finger follows the human hand motion.



a) Grasping a 100 mm diameter sphere with the human (left) and the augmented (right) hand.



b) Grasping a 100 mm edge cube with the human (left) and the augmented (right) hand.

Fig. 7. Grasping objects with the human hand and with the augmented hand.

contact forces are similar. Further details on the definition and evaluation of grasp matrix G are available in [19].

Fig. 7 shows some examples of grasps realized with the human and the augmented hand. The results, in terms of grasp quality matrix, are summarized in Table II. The Sixth–Finger significantly improved the grasp quality index, in particular for large size objects. These results confirm the intuitive observation that an additional robotic thumb

TABLE II

GRASP ISOTROPY INDEX, I.E. RATIO BETWEEN THE MINIMUM AND MAXIMUM SINGULAR VALUE OF GRASP MATRIX G, OBTAINED FOR DIFFERENT OBJECT SHAPES AND SIZES, WITH THE HUMAN AND THE AUGMENTED HAND.

Sphere	Human hand	Augmented hand
diam. 40 mm	0.0367	0.0404
diam. 60 mm	0.0302	0.0239
diam. 80 mm	0.0198	0.0212
diam. 100 mm	0.0137	0.0163
diam. 120 mm	0.0082	0.0134
diam. 140 mm	0.0060	0.0110
Cube	**Human hand**	**Augmented hand**
edge 40 mm	0.0334	0.0441
edge 60 mm	0.0226	0.0251
edge 80 mm	0.0109	0.0162
edge 100 mm	0.0064	0.0104
edge 120 mm	0.0040	0.0067
edge 140 mm	0.0025	0.0050

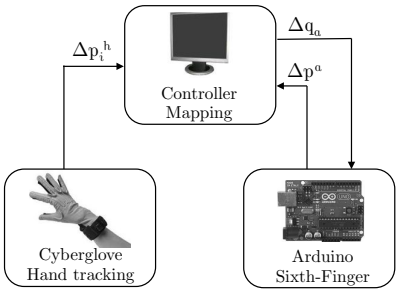


Fig. 8. Scheme of the Sixth–Finger control.

improves grasp capabilities of the human hand enlarging its workspace.

V. EXPERIMENTS

In order to evaluate the usability of the Sixth–Finger we grasped objects with different shapes using the proposed mapping algorithm to move the robotic device. A dataglove, the Cyberglove III System [26], was used to capture the motion of the human hand. We considered as reference points the fingertips of the human hand and the fingertip of the Sixth–Finger. The position of the reference points on the human hand are computed starting from the measures of the glove, using a direct kinematics algorithm. The reference point on the Sixth Finger was evaluated, in similar manner, from the measures of joint angles. Both the dataglove and the Arduino Uno board are connected through serial ports with a pc where the mapping algorithm runs. In Fig. 8 the scheme of the control approach is reported.

We grasped objects with different sizes and shapes. Some of the obtained grasps are shown in Fig. 9. Note that the box in Fig. 9-a and the dice in Fig. 9-c were not graspable using only the human hand fingers.

VI. CONCLUSION

In this paper we presented a first prototype of a wearable modular robot that can be used as an extra–finger. In particular, we developed a four DoFs device that can

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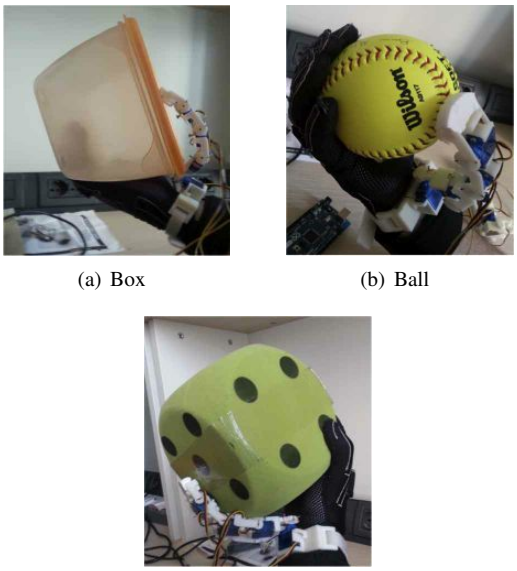

(a) Box (b) Ball

(c) Dice

Fig. 9. Different grasps performed with the aid of Sixth–Finger.

be worn on the user wrist by a rubber band. Such device plays the role of an extra thumb enlarging the human hand workspace and the hand dexterity. We also introduced a new mapping algorithm that is able to capture the human hand motion and generate the commands for the extra–finger. We provided several numerical and real experiments to prove the usability of the Sixth–Finger. Even if the study is in its preliminary phase, the first results, both numerical and experimental, show the feasibility and the usability of this type of device. Adding a supplementary robotic finger to the human hand, we have an improvement of its workspace size, that allows the user to grasp and manipulate larger objects. Furthermore we can enhance the range of tasks that can be performed by the hand. Concerning the design, the modular structure of the presented prototype is characterized by a simple and adaptable structure. We believe that the Sixth–Finger introduced in this paper could represent an interesting application in human–robot interaction and in particular in wearable robotics.

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